



Efgartigimod in the treatment of Guillain–Barré syndrome

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Abstract

Background Guillain–Barré Syndrome (GBS) is caused by immunoglobulin G (IgG) autoantibodies. Efgartigimod, a human IgG antibody Fc fragment that acts as a natural ligand for the FcRn, can increase IgG degradation, which thus may be a promising therapeutic drug for GBS.

Case presentation The two patients presented with postinfectious and acute flaccid paralysis. On admission, they were bedridden. Nerve conduction studies indicated peripheral neuropathy. GBS was suspected and they are treated with two doses of efgartigimod (10 mg/kg) within 5 days. Their muscle strength improved gradually and 4 weeks after the initial dose, they could walk independently. Following the first dose, Patient 1 complaint of muscle soreness, which subsided the next morning. Patient 2 was intubated due to respiratory failure the day after the initial dose, and did not report other adverse effects.

Discussion In GBS patients, two doses of efgartigimod (10 mg/kg) were effective in rapidly improving muscle strength, with a satisfactory safety profile. The findings suggest a potential role for efgartigimod in modifying the disease process in GBS patients.

Conclusion Efgartigimod seems effective and safe in the treatment of GBS. This study indicates the potential role of efgartigimod as a novel treatment option for GBS. Well-designed clinical trials should be conducted.

Keywords Guillain–Barré syndrome · Efgartigimod · FcRn inhibitor · IgG

Introduction

Guillain–Barré syndrome (GBS) is an immune-mediated polyneuropathy, with an annual global incidence of approximately 100,000 cases [1]. It typically presents as a monophasic disease marked by progressive flaccid paralysis. GBS is caused by immunoglobulin G (IgG) autoantibodies and molecular mimicry is the mechanism by which infectious agents trigger an immune response against autoantigens [2–4].

Current immunotherapies for GBS mainly include intravenous immunoglobulin (IVIg) and plasma exchange

(PLEX) [5]. IVIg is a biologic agent obtained through the fractionation of blood from 2000 to 16,000 healthy donors and plasma exchange is a blood separation technique [6]. IVIg and plasma are not always available, especially given the growing global shortage of blood products. In addition, it has been documented IVIg takes two weeks to become effective [7]. And patients treated with IVIg commonly experienced side effects (headache, tachycardia, nausea, vomiting, etc.), and 2–6% of patients had serious side effects (thrombosis, aseptic meningitis, renal insufficiency, neutropenia, etc.) [8]. Plasma exchange can show clinical effects within the first week of treatment [7], but it may also remove immune complexes and cytokines and can induce arrhythmias and hypotension [9]. Furthermore, despite the application of conventional immunotherapeutic approaches, around 3–7% of patients die, and 20% remain unable to walk after 6 months [10]. Therefore, there is an urgent need for effective, rapidly acting, and well-tolerated therapeutic drugs for GBS patients, independent of blood donations.

The neonatal Fc receptor (FcRn) ensures IgG homeostasis by preventing the degradation of IgG in lysosomes

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and prolonging their half-life [11]. Efgartigimod, a human IgG antibody Fc fragment that acts as a natural ligand for the FcRn, can increase IgG degradation [12]. In the Phase I trial of efgartigimod [13], it significantly reduced IgG levels without serious adverse events which provide a basis for treating IgG autoantibody-mediated diseases. Several studies have reported the safety and effectiveness of efgartigimod in IgG autoantibody-mediated diseases, such as myasthenia gravis, primary immune thrombocytopenia, pemphigus vulgaris, and foliaceous [14–18]. However, no study has yet investigated the efficacy of efgartigimod in patients with GBS. Here, we report the first two GBS patients treated with efgartigimod.

Case presentation

Patient 1

An 81-year-old female patient with a history of hypertension and diabetes mellitus presented with progressive limb weakness and numbness for 8 days. She also had a diarrhea bout 9 days prior to presentation. On admission, she was bedridden. The cranial nerves were intact. The Medical Research Council (MRC) scale values were grade 3 for neck flexors and bilateral upper limb muscles, and grade 2 for lower limb muscles. Deep tendon reflexes were decreased diffusely. Hypoesthesia to temperature and pain in a stocking-and-glove pattern was noted. Nerve conduction studies showed mixed demyelination and axonal sensory-motor abnormalities (Table 1). Cerebrospinal fluid (CSF) analysis revealed albuminocytological dissociation with a normal cell count and an elevated protein level of 1.56 g/L (reference range 0.2–0.45 g/L). She was diagnosed with GBS, even though serum anti-ganglioside antibodies (GM1, GM2, GM3, GM4, GD1a, GD1b, GD2, GD3, GT1a, GT1b, GQ1b, and sulfatide) were negative.

The patient was immediately managed with efgartigimod at a dose of 10 mg/kg, and the progression of her muscle weakness was halted. A second dose of efgartigimod (10 mg/kg) was administered 4 days later. After the initial dose, the patient had complaint of muscle soreness, which was untreated and subsided the next morning. The patient's condition improved gradually, as reflected by the GBS disability scale [19], Inflammatory Neuropathy Cause and Treatment (INCAT) disability score [20], MRC sum score [21, 22], Inflammatory Rasch-Built Overall Disability Scale (I-RODS) [23], and grip strength measured with an electronic hand dynamometer (EH101, CAMRY, CHINA) (Fig. 1a, c; Supplementary Table 1). The MRC sum score is the sum of the MRC score of six muscles (deltoid, biceps brachii, extensor carpi radialis, iliopsoas, quadriceps femoris, and the tibialis anterior) on both sides, with scores

Table 1 Nerve conduction study findings in the two patients

Never	Patient 1					Patient 2					
		CMAP (mV)/ SNAP (uV) ampli- tude	DML (ms)	MCV/SCV (m/s)	F wave latency (ms)	Frequency of F wave	CMAP (mV)/ SNAP (uV) ampli- tude	DML (ms)	MCV/SCV (m/s)	F wave latency (ms)	the fre- quency of F wave
Median	Left	1.32↓/NR	6.75↑	44.2↓/NR	29.5	60%↓	7.2/23.2	3.24	55.3/48.5	22.9	30%↓
	Right	2.1↓/NR	7.94↑	35.4↓/NR	28.8	90%	7.2/23.2	3.00	61.2/46.8	23.8	50%↓
Ulnar	Left	2.2↓/NR	4.20↑	49.5↓/NR	25.7	100%	6.1↓/10.1	2.44	61.0/47.9	24.9	70%↓
	Right	4.5↓/NR	3.05	41.0↓/NR	24.6	100%	6.1↓/8.5	2.38	63.2/46.2	25.7	100%
Tibial	Left	2.3↓/NR	6.17↑	NT/NR	NR	0%↓	5.0/ NR	3.28	NT/ NR	45.9	100%
	Right	1.86↓/NR	5.73↑	NT/NR	41.9	30%↓	5.7/ NR	4.12	NT/ NR	48.6	100%
common peroneal	Left	0.70↓/NR	6.72↑	27.8↓/NR	67.4↑	20%↓	4.4/2.3	3.51	49.5/48.7	46.1	40%↓
	Right	0.27↓/NR	11.9↑	34.2↓/NR	NR	0%↓	6.0/1.42	3.68	45.6/51.3	45.6	50%↓

CMAP compound motor action potential; SNAP sensory nerve action potential; DML distal motor latency; MCV motor conduction velocity; SCV sensory conduction velocity; ↑ increase; ↓ decrease; ↑ increase; ↓ decrease; NT not tested; NR non-responsive

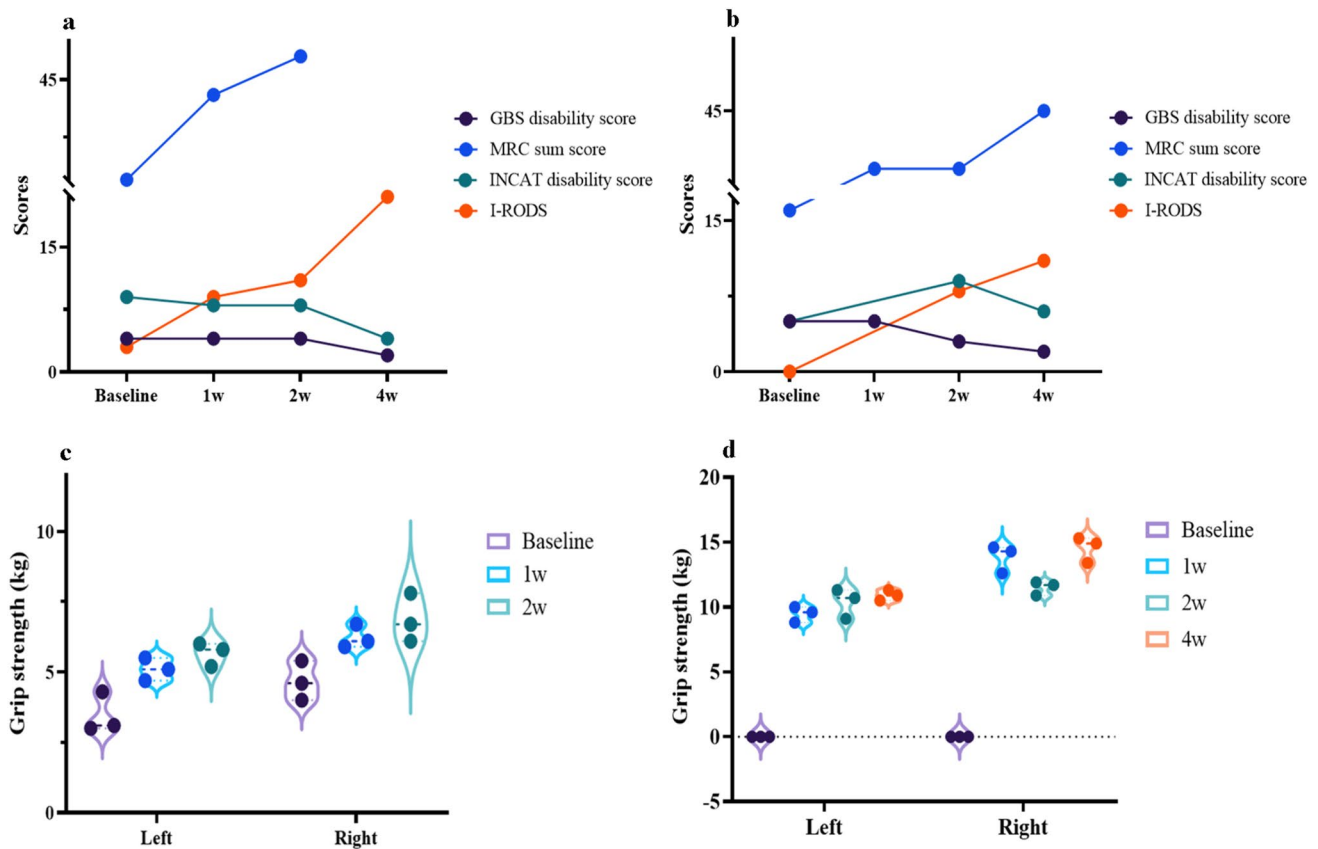


Fig. 1 The changes of the GBS disability score, Inflammatory Neuropathy Cause and Treatment (INCAT) disability score, Medical Research Council (MRC) sum score, Inflammatory Rasch-Built Overall Disability Scale (I-RODS), and grip strength in Patient 1 (**a, c**) and Patient 2 (**b, d**)

ranging from 0 to 60 [21, 22]. Two weeks after the initial dose, the serum albumin level (reference range 40–55 g/L) decreased by 8.93%, and the serum IgG level (reference range 8.6–17.4 g/L) decreased by 52.21% (Fig. 2; Supplementary Table 1). Four weeks later, the patient was able to walk independently.

Patient 2

A 65-year-old woman with a history of hypertension and coronary heart disease presented with progressive weakness and numbness in four limbs for 3 days and shortness of breath for 2 days. She had a sore throat and stuffy nose 7 days earlier. Physical examination on admission revealed bilateral ptosis and dysarthria. The MRC scale value was grade 1 for 4 limb muscles and grade 2 for neck flexors. Deep tendon reflexes were diminished diffusely. Hypoesthesia to temperature and pain in hands and feet were noted. A nerve conduction study revealed reduced amplitudes of bilateral ulnar nerve in motor conduction (Table 1). Arterial blood gas analysis showed a low PO_2 (55 mmHg) and a high PCO_2 (46.6 mmHg). A diagnosis of GBS was made,

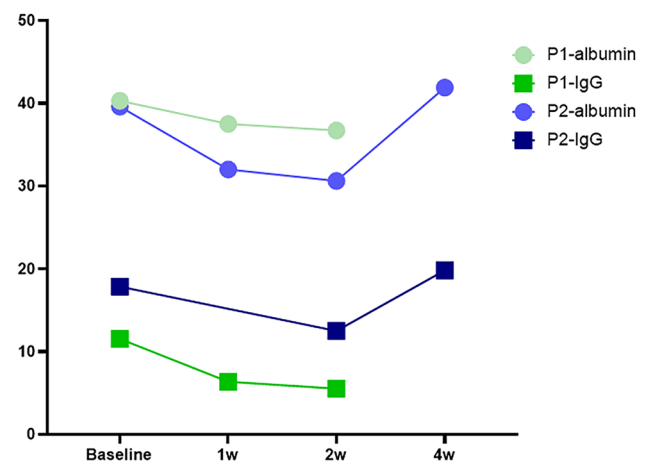


Fig. 2 The levels of albumin and IgG at baseline and 1 week, 2 weeks, and 4 weeks after the initial dose of efgartigimod in the two patients

although lumbar puncture and anti-ganglioside antibody analysis were not conducted.

The patient was treated with low-flow oxygen therapy and a dose of efgartigimod (10 mg/kg). Three hours later, her

muscle strength improved dramatically with an MRC scale value of 3 in four limbs. However, the patient was intubated the next morning for respiratory failure and transferred to the intensive care unit (ICU). On the same day, another dose of efgartigimod (10 mg/kg) was administered. A gradual improvement of muscle strength was noted (Fig. 1b, d; Supplementary Table 1) and she was successfully extubated after 12 days. Two weeks after the initial dose, the serum albumin level decreased by 22.72%, and the serum IgG level decreased by 29.97% (Fig. 2; Supplementary Table 1). Four weeks later, the patient was able to walk independently.

Discussion

GBS is considered an IgG antibody-mediated autoimmune disease [3]. Studies support that blockade of the FcRn may serve as a unique treatment approach for such diseases [24]. In the present study, we reported two GBS patients, who were treated with a neonatal Fc receptor antagonist efgartigimod and experienced rapid improvement within 4 weeks.

The fundamental mechanism underlying FcRn-mediated IgG recycling is its pH-dependent affinity [25, 26]. When IgG undergoes endocytosis and enters cells within the acidic environment of endocytic vesicles (pH 5.0–6.5), FcRn binds to IgG, preventing its transport to lysosomes for enzymatic degradation. In the extracellular physiological environment (pH 7.4), their affinity diminishes, leading to the release of IgG and completing the recycling process. Efgartigimod has been designed with ABDEG-equipped molecules to increase affinity while preserving the pH-dependent characteristics of FcRn interaction to decrease pathogenic IgG [12]. A study suggested that efgartigimod has a more effective and rapid impact on clearing IgG in cynomolgus monkeys compared to standard IVIg therapy [13].

To date, there have been no published clinical studies on the treatment of GBS patients with efgartigimod. The Phase I clinical trial indicated that multiple injections of 10 mg/kg efgartigimod resulted in the maximum decrease in IgG levels. Higher doses did not significantly lower IgG levels further and increased the risk of adverse events [13]. Therefore, two doses of efgartigimod (10 mg/kg) were administered for our patients in consideration of the monophasic course of GBS, and they responded well to the treatment. Four weeks after the initial dose, the score of the GBS disability scale decreased by 2 and 3 in the two patients, respectively, and the mean change of the score was less than 2 in GBS patients treated with IVIg or PLEX [27]. However, the best regimen for efgartigimod should be investigated in the future.

Efgartigimod is well tolerated in our two patients. Patient 1 experienced muscle soreness after treatment. In fact, myalgia is common in GBS patients [28] and might be not associated with efgartigimod. Patient 2 did not

report adverse effects, and respiratory failure following intravenous infusion of efgartigimod was considered as a manifestation of disease progression.

The FcRn plays a crucial role in mediating the recycling pathways of both IgG and albumin, yet IgG interacting at a distinct site from albumin [29]. Efgartigimod has selectivity for the IgG binding site of FcRn, and it did not reduce albumin levels in patients with myasthenia gravis [15]. However, Patient 2 exhibited a decrease in albumin levels two weeks after the initial dose. As the median serum albumin level also decreased in IVIg-treated GBS patients [30], the causal link between efgartigimod and decreased albumin levels in GBS patients remains unclear and further studies are needed.

This study pioneers in presenting case reports of GBS patients undergoing treatment with efgartigimod. It provides detailed documentation of drug dosage, drug usage, and changes in relevant scales post-treatment. Additionally, attention is given to the variations in IgG and albumin levels. Encouragingly, both patients achieved favorable outcomes, bolstering confidence in using efgartigimod for treating GBS and other IgG-mediated immune-related disorders.

Conclusion

In conclusion, efgartigimod with two doses of 10 mg/kg demonstrates rapid and significant neurological relief in the treatment of GBS, with good tolerability. The findings suggest that efgartigimod may be a promising new treatment option for GBS, warranting further investigation in studies with larger sample sizes.

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Author contributions All authors contributed to the study conception and design. Wei Zhang and Huiqiu Zhang wrote the first draft of the manuscript. All authors read and approved the manuscript.

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Data availability The data underlying this article are available in the article and in its online supplementary material.

Declarations

Conflicts of interest None of the authors have any financial and conflict of interest to disclose.

Ethical approval This report has been performed in accordance with the ethical standards laid down in the 1964 Declaration of Helsinki and its later amendments. All the patients provided written informed consent according to the Declaration of Helsinki.

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